How Mkoba Works Today

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How each Mkoba feature behaves in the current code (not as planned). Each feature follows the same pattern: **steps** (what happens, in order), **rules** (limits and calculations), **differences** from similar features where they matter, and **weak points**. Commands are shown in Swahili; most also have an English form.

1. Sign-up and consent

Steps

1. A new number messages Mkoba. Mkoba creates a vendor row for that phone straight away (name = the phone number) and stores the raw message.
2. Whatever the first message said, the reply is the welcome with the data notice and a link to `mkoba.digital/faragha` (or `/privacy`). The first message itself is not acted on.
3. Until the vendor agrees, every text gets a reminder and nothing else is processed. Any of `nakubali`, `kubali`, `ndio`, `yes`, `i agree`, `accept` counts as agreeing. Only `FUTA DATA` works before consent.
4. After agreeing: Mkoba asks for the name. A reply with no digits, up to 4 words, not a greeting and not a command is taken as the name ("naitwa", "my name is" are stripped).
5. Then it asks for the type of business (up to 6 words, no digits). `ruka` / `skip` ends sign-up at either question.
6. Done: Mkoba sends the help list and invites the vendor to set their stock.

Rules

- Language for the welcome follows the first message (English words → English).
- A sign-up can also happen on USSD (agree, then name).

Differences

- Riders (`boda`), households (`nyumbani`) and shop customers (`duka 4108`) never go through vendor sign-up.
- Someone who has ordered from a shop before gets a shop hint instead of sign-up, unless they send `sajili`.

Weak points

- The vendor row and raw message are stored **before** consent.
- If the name answer isn't recognised (for example it is a record like `nimeuza chips 5000`), the message is handled as a record and the vendor's name stays as their phone number until they answer correctly.
- `ndio` counts as consent, so a vendor answering an unrelated question can agree without reading.

2. Recording sales, purchases and expenses

Steps

1. The vendor sends a message, for example `nimeuza chips 5000` or `nimenunua mchele kilo 10 kwa 25000`.
2. Mkoba splits it into one record per line, and again at each action word (`nimeuza ... nimenunua ...` gives two).
3. For each record it finds the **kind** from the action word, the **amount**, the **quantity and unit** (kilo, lita, gramu, vipande/pcs), and the **item** (the words left over).
4. It scores how sure it is:
   - sure (one amount marked as money, such as `5,000`, `5k`, `elfu tano`, or one plain number) → **saved at once**, reply `Sawa! Nimeongeza ...`;
   - fairly sure (several plain numbers) → saved as **pending**, and Mkoba asks `NDIO` / `HAPANA`;
   - not sure (no action word, no amount, two kinds of action word, two money amounts) → **nothing saved**, reply `Samahani, sikuelewa...`, and the message is logged for review.
5. After saving, the reply adds stock warnings (section 8) and a bulk-buying hint if the vendor's group is buying that item (section 11).

Rules

- **Kinds:** sale (`nimeuza`, `nauza`, `mauzo`, `nimepokea`, `nimelipwa`, `sold`), purchase (`nimenunua`, `manunuzi`, `bought`), expense (`nimelipa`, `nimetumia`, `gharama`, `paid`, `spent`).
- **Typos** in action words are corrected (`nmeuza`, `nimeuuza` → `nimeuza`), but real words that look similar (`nimeuliza`, `nimekopa`) are left alone.
- **Number words:** Swahili (`elfu kumi na tano`, `laki moja`) and slang (`buku tano`, `jero mbili`). No English number words.

- **Price memory:** a plain number under 100 after an item (`nimeuza soda 10`) is a count. Mkoba multiplies by the last unit price for that item and always asks NDIO/HAPANA. With no earlier price it asks `soda 10 kwa bei gani?` and saves nothing.
- **Pending records:** a new record replaces any unanswered one; NDIO/HAPANA works for **1 hour**.
- **Date:** a record is dated when Mkoba receives it. There is no way to back-date.
- Records from a **voice note** always wait for NDIO.

Differences

- Debt messages (`Juma anadaiwa 5000`), pasted mobile-money SMS, and every command are checked **before** this step, so they are never saved as ordinary records.
- Receipts (section 9) and USSD (section 19) save sales without asking.

Weak points

- `nimelipa Juma 2000` is an expense; only `nimemlipa` is a debt payment.
- A single plain number is saved without asking, even if it was meant as a count of 100 or more.
- No back-dating: a record sent the next morning counts for the wrong day.
- The item is whatever words are left, so `nimeuza chips za mayai 5000` saves the item as "chips za mayai" (fine), but extra words like a customer's name end up in the item.

3. Undo and correct

Steps

1. `futa` (also `ondoa`, `futa ya mwisho`, `undo`, `delete`): Mkoba finds the newest saved record, then every saved record that came from the **same message**, and deletes them all. Reply: `Nimefuta ...`.
2. Sending `futa` again deletes the message before that, and so on.
3. `rekebisha 6500` (also `badilisha`, `sahihisha`, `correct`, `edit`): changes the amount of the newest saved record. Reply: `Nimerekebisha ...: TSh 6,000 → TSh 6,500.`

Rules

- Deleting is a soft delete: the row is marked deleted and drops out of every total, but stays in the database (until full data deletion).
- `rekebisha` refuses when the last message produced several records; the vendor must `futa` and resend.

- `rekebisha` with no amount asks for one.
- Pending (unanswered NDIO) records are ignored by both.
- Every correction is logged to help improve the parser.

Differences

- Both act on **any** kind of saved record, not just sales: a debt, a receipt sale, a pasted mobile-money record, a completed order, or a bulk purchase. Whichever was saved last is the one changed.
- USSD option 8 deletes only the single newest record, not the whole message.

Weak points

- No way to pick an older record ("delete yesterday's chips"); only the newest, one message at a time.
- `futa` on a receipt deletes the sale, so the customer's points disappear, but the receipt number stays used and the customer still has the forwarded receipt.
- `futa` after `maliza oda` deletes the order's sale records (stock goes back, since stock is recalculated from records), but the order stays marked as completed.
- No way to change the kind or item, only the amount.

4. Summaries

Steps

1. The vendor sends a period word: `leo` / `today` (also `muhtasari`, `ripoti`), `jana` / `yesterday`, `wiki` / `week`, `mwezi` / `month`, `salio` / `balance` (also `jumla`, `total`).
2. Mkoba adds up saved records for that period: sales, purchases, expenses, each with a count, and **Faida** (profit) or **Hasara** (loss).
3. For every period except `salio` it adds: best seller, biggest cost, and the profit or loss of the period before (yesterday, last week, last month).

Rules

- Profit = sales − purchases − expenses. Debts are not counted.
- Periods use Tanzania time. The week starts Monday.
- Only confirmed, undeleted records count; pending NDIO records don't.
- `msaada` / `help` / `menu` sends the command list instead.

Differences

- USSD gives today and this week only, without the best seller and comparison lines.
- The dashboard and app show the same totals plus a 14-day chart.

Weak points

- Purchases count fully as a cost on the day they are bought, so a big stock purchase shows a loss that day even if the goods sell later.
- No custom date range (for example "last March").

5. Evening and weekly reports

Steps

1. A server timer runs `node reports.js` at **21:00** Tanzania time every day.
2. Monday–Saturday it reports **today**; Sunday it reports **this week**.
3. It picks vendors who agreed, have reports on, and saved at least one record in that period. Others are skipped.
4. For each vendor it first claims (vendor, period, date) in `report_log`, so a second run the same day sends nothing twice.
5. If the vendor messaged within the last 23 hours, it sends a full text: greeting, the summary with best seller and comparison, on Sunday the top 5 people who owe them, any low-stock items, and "to stop, send ACHA RIPOTI".
6. Otherwise it sends the approved WhatsApp template `mkoba_muhtasari` with name, period, sales, purchases, expenses and profit only.
7. If sending fails, the claim is removed so a later run can retry.
8. The same run then closes due bulk orders (Friday), saves credit-score snapshots, and re-checks pending payments.

Rules

- `ACHA RIPOTI` / `STOP REPORTS` turns reports off; `ANZA RIPOTI` / `START REPORTS` turns them on. On is the default.
- The template is always in Swahili.

Weak points

- **Bug:** the 23-hour check counts any logged activity, including Lite App and USSD use. A vendor who used only the app or USSD today is treated as "in the window", the plain text fails because WhatsApp refuses it, and no template is sent instead. They get no report.

- The template only works once Meta approves `mkoba_muhtasari`; until then out-of-window vendors get nothing.
- English vendors get the Swahili template outside the window.
- One run at 21:00: if the server is down then, that day's report is not sent later.

6. Debts and reminders

Steps

1. The vendor sends a debt message. Mkoba recognises four kinds:
   - they owe me: `Juma anadaiwa 5000`, `namdai Juma 5000`, `nimemkopesha Juma 5000`, `Juma owes me 5000`, `lent Juma 5000`;
   - I owe them: `Juma ananidai 5000`, `nadaiwa na Juma 5000`, `nimekopa kwa Juma 5000`, `I owe Juma 5000`;
   - they paid me: `Juma amelipa 2000`, `Juma amenilipa 2000`, `Juma paid 2000`;
   - I paid them: `nimemlipa Juma 2000`, `nimelipa deni la Juma 2000`, `nimemrudishia Juma 2000`.
2. It is saved **at once** (no NDIO), with the person's name, and the reply shows the new balance with that person.
3. Without an amount, Mkoba asks for one and saves nothing.
4. `madeni` lists everyone with a non-zero balance in two groups, **they owe you** and **you owe**, largest first, with days since the last debt was added.
5. `deni Juma` shows Juma's balance. If Juma owes the vendor, Mkoba also sends a ready-made reminder ("Habari Juma, hii ni kumbukumbu ya deni lako … TSh … (tangu …)") for the vendor to forward.

Rules

- The name is one word, or a title plus one word (`Mama`, `Baba`, `Bi`, `Bwana`, `Kaka`, `Dada`, `Mzee`, `Ndugu`, `Shangazi`, `Mjomba`, `Mr`, `Mrs`). Names are matched without regard to capitals.
- Debt messages are checked **before** ordinary records, so `Juma amelipa 2000` is never an expense.

Differences

- Debts do **not** count as sales or expenses in profit. Lending goods on credit, and being paid back, change only the debt balance.

- The reminder is forwarded by the vendor, like a receipt; Mkoba never messages the customer.
- The Sunday report lists the top 5 people who owe the vendor.

Weak points

- Goods sold on credit never appear as a sale, not when given and not when paid, so sales and profit are understated for vendors who sell on credit.
- `nimelipa Juma 2000` (without the **m**) is saved as an expense.
- `Juma Ali anadaiwa 5000` saves under "Ali"; two spellings of one name become two people.
- No due dates and no automatic reminders; the vendor has to ask.

7. Mobile money SMS paste

Steps

1. The vendor copies a mobile-money confirmation SMS and pastes it into the Mkoba chat.
2. Mkoba matches it against known SMS formats:
   - **paid a business or till** (`Paid ... Tsh to ... Till ...` , `You have paid ... to ...` , `Payment to ... is successful`) → a **purchase** for the amount, plus an **expense** for the fee;
   - **bundle** (`You have paid ... for ... BUNDLE`) → an **expense**;
   - **sent to a person** (`Paid ... TSh to 2557... NAME. Charges ...` , `SENT TSh ... to ...`) → Mkoba asks: 1 purchase, 2 business expense, 3 not business;
   - **not business money** (agent cash deposit, transfers to own card, bank-card SMS, Kenyan shillings) → not recorded, with a reason.
3. Purchases and expenses are saved as **pending** and Mkoba asks `NDIO` / `HAPANA` .
4. For a person transfer, the answer `1` / `2` saves it as a purchase or expense (the fee is always an expense); `3` discards it. `ndio` re-asks the question.

Rules

- The transaction ID (TID) is stored; pasting the same SMS twice is refused as a duplicate.
- Pending SMS records follow the same 1-hour rule as other pending records.
- A pasted SMS never changes the vendor's language.

Weak points

- Only **Airtel Money Tanzania** formats are recognised. M-Pesa Tanzania, Mixx by Yas (Tigo Pesa) and HaloPesa SMS are not, and fall through to the normal record parser (usually "sikuelewa").
- **Money received** (a customer paying the vendor) is not supported yet, which is the most useful case for sales.
- Any payment to a business is assumed to be a purchase of stock; paying a till for rent must be fixed by hand.
- Formats were built from a small number of real samples; any change in the provider's wording breaks recognition.

8. Stock

Steps

1. Stock is **calculated**, not stored: for each item and unit, Mkoba takes the vendor's last count (if any), then adds purchases and subtracts sales recorded **with a quantity** after that count.
2. The first time an item moves with a quantity, the reply invites the vendor once to enter what is on the shelf: `stoo mchele kilo 20`.
3. `stoo mchele kilo 20` (or several: `stoo mafuta lita 5, sukari kilo 2`) saves a count; later records are added to or taken from it.
4. `stoo` lists every item with its quantity, ⚠️ for low items, and roughly how many days it will last. Items with an unknown amount show `?` and an example.
5. When a sale pushes an item to or below its alert level, the sale's reply adds `⚠️ Stoo ya ... inakaribia kuisha`; when it reaches zero, `imeisha`.
6. The evening report lists low items.

Rules

- **Alert level:** the vendor's own (`stoo mchele chini ya 5`), or automatic: about **2 days** of sales, based on the last 14 days. Automatic alerts start only after **3 sales with a quantity** in 14 days. `chini ya auto` returns to automatic.
- An alert fires once, when the level is **crossed**, not on every sale below it.
- An amount is known only if the vendor counted it or bought it with a quantity; sales alone give `?`.
- Units: kg, l, g, and pieces (no unit). The same item in two units is two separate lines.

Differences

- Completed shop orders and `jirani imefika` (bulk goods received) change stock too, because they create records with quantities.
- Records without a quantity (`nimeuza chips 5000`) never change stock.

Weak points

- Item names must match exactly: `mchele` and `mchele mweupe` (or `rice`) are different items.
- Most food vendors sell by amount, not quantity, so for them stock rarely moves.
- No unit conversion (a 25 kg bag bought, sold by the kilo, works only if both are recorded in kg).
- Spoilage, own use and theft are invisible until the vendor recounts.

9. Receipts and points

Steps

1. The vendor sends `risiti 0712345678 chips 5000`. Several items can go on separate lines; a line without an action word is treated as a sale.
2. Mkoba saves the sale(s) **immediately, without NDIO/HAPANA**. If any line can't be read as a sale with an amount, nothing is saved and the vendor gets the usage hint.
3. It gives the receipt the vendor's next number (R-0001, R-0002…), with no gaps or repeats, linked to the customer's phone.
4. The vendor gets two messages:
   - a confirmation: sale amount, receipt number, customer number, points earned and total, any stock warning, and "send the next message to the customer: press and hold it, then Forward";
   - the receipt: shop name, date and time, items, total, points earned and total, and "Thank you".
5. The vendor forwards the receipt; it reaches the customer from the vendor's own WhatsApp number.

Rules

- 1 point per full TSh 1,000 of sales on the receipt.
- Points are calculated from the live sales on each receipt, so `futa` on that sale removes its points.

- `pointi 0712345678` shows one customer; `pointi` the top 10.
- `tumia pointi 0712345678 20` uses points (refused if the customer has fewer) and gives the vendor a message to forward to the customer.

Differences

- `maliza oda` (shop orders) sends the receipt to the customer directly if they messaged Mkoba in the last 24 hours; otherwise the vendor forwards it.

Weak points

- Delivery depends on the vendor forwarding; Mkoba can't know if they did.
- A mistyped customer number still creates the receipt and gives points to the wrong number.
- Receipt sales skip the NDIO check.
- Rewards are not defined in the system; the vendor decides and tracks them by hand.

10. Shop catalog and orders

Steps (vendor sets up)

1. `bidhaa ongeza chips 3000, soda 600` adds items or updates prices. The first time, the vendor gets a shop code (for example 4108).
2. `bidhaa` shows the numbered list, the shop code, the web link `mkoba.digital/k/4108` and a WhatsApp link that opens Mkoba with `duka 4108` typed in.
3. `bidhaa imeisha chips` / `bidhaa ipo chips` hides or shows an item; `bidhaa ondoa chips` deletes it. An item can be named or numbered.

Steps (customer orders on WhatsApp)

1. The customer sends `duka 4108` (or `shop 4108` for English). Mkoba sends the menu and a privacy line, and opens a **2-hour** shopping session.
2. They pick by number: `1x2, 3` (item 1 twice, item 3 once). Mkoba replies with the cart. Other words: `kikapu` (cart), `menyu`, `futa kikapu` (clear), `toka` (leave).
3. `tuma` / `agiza` / `send` asks for delivery details. The customer replies `name, place`, or `chukua` (pick up), or shares a location pin.
4. The order is created with the next order number for that vendor. The customer gets `Oda #... imetumwa`.

Steps (customer orders on the web)

1. The customer opens `mkoba.digital/k/4108`, sets quantities, enters name, phone, and pick-up or a delivery place.
2. The order is created; the page offers **Lipa sasa kwa simu** (pay now), or pay on receipt.

Steps (vendor handles the order)

1. If the vendor messaged Mkoba in the last 23 hours, they get the order at once; otherwise it is shown with their next message.
2. `oda` lists open orders; `oda 1` shows one.
3. `kubali oda 1`: the customer is told (if within their window); a delivery order gets a boda job (section 16); if payments are on, the customer gets a payment link for goods plus delivery.
4. `kataa oda 1`: the customer is told; a pending payment is cancelled; a paid one is flagged "refund needed" for the admin.
5. `maliza oda 1`: records one sale per line (with quantities, so stock moves), creates a receipt and points, and sends the receipt to the customer or to the vendor to forward.

Rules

- Up to 99 of each item. Web orders: at most 5 per 10 minutes per internet address, plus a hidden anti-bot field.
- `maliza` works on new or accepted orders; a closed order can't be reopened.

Weak points

- The vendor learns of an order only within WhatsApp's window or on their next message; an order can wait unseen for days.
- Web orders accept **any phone number** without checking it. A wrong or someone else's number gets the order, and that person is later treated as a past customer (offered the shop instead of sign-up unless they send `SAJILI`).
- The web shop is Swahili only.
- `maliza oda` does not check payment; the vendor can complete an unpaid order.
- Ordering doesn't check stock, only the item's available/sold-out flag.

11. Bulk buying groups

Steps

1. **Groups are collection points.** The admin creates them on `/admin/bulk`, or a vendor sends `unda kikundi Sinza Mori, Sinza` (name, area) and joins it automatically. Each gets a code such as `SIN01`.
2. **Joining:** `vikundi` lists the 20 largest groups; `jiunge SIN01` or `jiunge Sinza` joins by code, name or area (several matches → Mkoba lists them). Sharing a location saves it and **suggests** the nearest group within **3 km**.
3. **Hints:** after a vendor saves a purchase, Mkoba may suggest the group order (if the group or admin price list has that item) or suggest joining a group. At most once a week.
4. **Ordering:** `jirani mchele kilo 20` (or several items) adds to the group's open order for the week, created on first use. Sending the same item again replaces the quantity. `jirani` shows the group total per item, members, the admin's price, and the vendor's share. `jirani ondoa mchele` removes it.
5. **Closing:** the order closes **Friday 20:00**; the 21:00 evening run sets each line's price from the admin price list and sends every member their share, price, total and where to collect. If payments are on, the message includes a payment link for the priced part.
6. **Goods arrive:** the admin presses **Goods arrived** on `/admin/bulk`. Members who collect at the point are told to collect and send `jirani imefika`; members with home or place delivery get boda jobs (section 16).
7. **Received:** `jirani imefika` saves the vendor's priced share as **purchases** with quantities (so stock goes up). Items without a price are listed for the vendor to record themselves.

Rules

- Where to receive: `pokea nyumbani <address>`, `pokea <place>`, or `pokea kikundi` (default). `pokea` shows the current choice.
- `toka kikundi` leaves the group and removes the vendor's lines from the open order.
- Messages outside WhatsApp's window are kept and delivered with the vendor's next message.

Weak points

- Grouping is **suggested**, not automatic: nobody is placed in a group without `jiunge`, and orders are not merged across nearby groups.
- Prices come only from the admin's price list; there is no wholesaler, minimum order, or price tier by volume.
- Buying, transport and splitting the goods are done by hand outside the system; the only system step is the Goods arrived button.

- A member who doesn't pay or doesn't collect isn't handled.
- Item names must match the price list exactly to get a price.
- Closing time is fixed in code (Friday 20:00).

12. Households

Steps

1. A new number sends `nyumbani` (also `kaya`, `familia`, `household`, `for my home`). Mkoba registers it as a **household** and sends a household data notice.
2. The household replies `NAKUBALI`; Mkoba asks for a name.
3. Mkoba sends 3 steps: join a group (`vikundi`, `jiunge SIN01`, or share location), choose where to receive (`pokea nyumbani <address>`, `pokea <place>`, `pokea kikundi`), then order (`jirani mchele kilo 5`).
4. From here everything works as in section 11: same groups, same weekly order, same Friday close, same payment link.
5. `jirani imefika` only marks the goods received; nothing is recorded as a purchase.

Rules

- A household **must** choose where to receive before it can add to an order.
- Only bulk-buying commands, `LIPA`, the language switch and `FUTA DATA` work; anything else gets the 3-step help.
- Households get no evening reports.

Differences

- Stored in the vendors table with kind = household, so a household and a vendor share groups and orders.
- The only real difference from a vendor in a group is the receiving point and that no records are kept.

Weak points

- One phone can be a vendor **or** a household, not both. A vendor who sends `nyumbani` simply stays a vendor.
- A household that doesn't send a valid name at step 2 keeps its phone number as its name, without being asked again.
- Home delivery depends on the boda feature and on the admin pressing Goods arrived.

13. Credit score

Steps

1. The vendor sends `alama` (or `score`).
2. Mkoba calculates the score from the last 90 days of saved sales, purchases and expenses, plus debts.
3. If there isn't enough history, it says how many record days the vendor has and how many are needed.
4. Otherwise it replies with the score out of 100, a grade (A–E), the six parts with their points, tips for the two weakest parts, and a note that the score appears on the business statement.
5. A snapshot is saved (one per day); the 21:00 run also saves snapshots for vendors who have lender consents.

Rules

Part	Points	Full points when
Recording every day	30	records on every day since the first record (window 30–90 days)
Profit	25	profit in each of the last three 30-day months (15) and a 30% margin (10)
Sales trend	15	last 30 days' sales ⩾ 1.2× the earlier average; neutral 7.5 with under 15 days of earlier history
Time using Mkoba	15	180 days since the first record
Debts	10	owes nothing; 0 if owing a month of sales or more
Verified records	5	30% of records from mobile-money SMS, completed orders, bulk deliveries or receipts

- Needs at least **14 days with records** and **30 days** since the first record.
- Grades: A ⩾ 80, B ⩾ 65, C ⩾ 50, D ⩾ 35, E below.

Weak points

- The weights are reasoned guesses, never tested against real loan repayment.
- Almost everything is typed by the vendor; inflating sales or skipping costs raises the score.

- "Verified" is weak: a pasted SMS is just text the vendor could edit, a receipt is typed by the vendor, and a shop order could be placed by the vendor themselves.
- Goods sold on credit don't count as sales (section 6), which lowers the profit part for credit sellers.
- The time-using-Mkoba part rewards age, not behaviour.

14. Business statement

Steps

1. The vendor sends `taarifa ya biashara` (also `ripoti ya biashara`, `taarifa ya mkopo`, `statement`).
2. Mkoba **closes every older statement link** for that vendor, then creates a new secret link `mkoba.digital/s/...` valid for **30 days**.
3. The vendor sends the link to a lender or SACCO.
4. Opening the link shows a printable page (Swahili, with an English switch): business name, phone, using Mkoba since, a 6-month table (sales, costs, profit, days with records), monthly averages, share of days with records, money owed to and by the business, share of records from mobile-money SMS, and the credit score with its parts.
5. `funga taarifa` / `close statement` closes all links at once.

Rules

- Only a hash of the link is stored; a lost link can't be recovered, only replaced.
- Expired or closed links show "this link has expired or was closed by the business owner".
- The page asks search engines not to index it.
- The page states that the figures are the vendor's own records and that Mkoba has not verified them.

Differences

- A **connected lender** (section 15) sees the same figures in their own portal, without a link, for as long as the vendor allows.

Weak points

- Anyone who has the link can open it for 30 days; a forwarded link spreads with no record of who opened it.
- Asking for a new statement silently kills the link already sent to another lender.

- The page says SMS records were "taken directly from the provider's messages", but they are text the vendor pasted and could have edited. That sentence overstates their reliability.

15. Lenders

Steps

1. The admin adds a lender on `/admin/lenders` (code such as `NMB`, name) and gets a generated password to give them.
2. The lender logs in at `mkoba.digital/l` and requests access by entering a vendor's phone number, with an optional note.
3. If that number is a vendor, the vendor gets the request on WhatsApp (now, or with their next message if outside the window): who is asking, what they will see, and the commands.
4. The vendor replies `ruhusu NMB` (or `ndio NMB`) to allow, `kataa NMB` (or `hapana NMB`) to decline. A vendor can also allow a lender who never asked.
5. With access, the lender's portal lists the vendor with their score history and the same statement as section 14. Every view is logged.
6. `wakopeshaji` shows the vendor every lender, their status, and when they last looked. `ondoa ruhusa NMB` removes access at once.

Rules

- Every request, grant, decline and removal is stored with who did it.
- A lender can send up to 30 requests a day; login is limited to 5 tries per 15 minutes; sessions last 12 hours.
- The portal does not say whether a number uses Mkoba (it shows the same "request sent" either way).
- Passwords are stored hashed (scrypt); the admin can reset or switch off a lender.
- Daily score snapshots are saved for vendors who allowed at least one lender.

Weak points

- Access has no end date; it lasts until the vendor removes it.
- One shared password per lender organisation, no individual staff accounts, no two-step login.

- `ndio NMB` counts as allowing, close to an ordinary YES; a vendor could grant access without reading.
- No lender is connected yet, so the flow is untested with a real lender.

16. Boda delivery

Steps (riders join)

1. The admin adds a rider on `/admin/riders`, or a new number sends `boda` (also `bodaboda`, `dereva`, `rider`), gives a full name and work area, and waits.
2. The admin approves the rider; the rider is told and can start.

Steps (a job is created)

1. **Shop order:** when the vendor sends `kubali oda` for an order with delivery, a job is created from the shop to the customer.
2. **Bulk share:** when the admin presses Goods arrived, a job is created for each household or vendor who chose home or a named place.
3. The fee is calculated (see Rules). The customer or household is told the fee; for bulk shares a payment link is sent, otherwise they pay the rider cash.
4. Available riders within **5 km** of the pickup are told about the job, but only those inside WhatsApp's 24-hour window.

Steps (rider works)

1. `kazi` lists up to 10 open jobs, nearest first (if the rider shared a location).
2. `chukua 12` takes job 12. The rider gets pickup and drop-off details with phone numbers, and is warned not to take cash if it was paid by phone. The vendor and customer get the rider's name and phone.
3. `nimechukua 12`: the customer is told it is on the way. `nimefikisha 12`: the vendor is told `Oda #... imefikishwa`.
4. `acha 12` gives the job back and it is offered to riders again. `zima` / `washa` turns new-job messages off and on.

Rules

- **Fee** = TSh 1,000 + TSh 750 × (straight-line km × 1.3), rounded up to the next 500, minimum 1,500. All values can be changed on `/admin/riders`.
- Without a map position at both ends, the fee is "to agree with the rider". The vendor is reminded to share the shop's location.

- A rider can hold at most 3 unfinished jobs.

Weak points

- Straight-line distance × 1.3 is a rough road estimate; real routes can be much longer.
- Riders outside the 24-hour window aren't told about new jobs; they must check `kazi`.
- `nimefikisha` is the rider's word only; the customer never confirms delivery.
- No rider rating, no proof of delivery, and riders are paid by the admin by hand.
- A named place (not home) has no map position, so its fee is always "to agree".

17. Payments (Selcom)

Steps

1. Mkoba creates a payment in three cases:
   - **shop order:** when the vendor accepts it (goods plus delivery fee); a web customer is offered payment for the goods right after ordering;
   - **bulk share:** when the weekly order closes (the priced part of the member's share);
   - **delivery fee:** for home or place delivery of a bulk share.
2. Each payment gets a reference (`MKB...`), a Selcom checkout order, and a short link `mkoba.digital/pay/MKB...`. The payer gets `💳 Lipa ... kwa simu: andika LIPA ..., au fungua: <link>`.
3. The payer either sends `LIPA` (Mkoba asks Selcom to send a PIN prompt to the phone for their newest unpaid payment) or opens the link and pays on Selcom's page.
4. Mkoba learns the result from Selcom's notification, from the payer opening the link again, from `LIPA`, or from the 21:00 check. In every case it **asks Selcom for the status** rather than trusting the notification.
5. On payment: the payer gets `✅ Malipo ya ... yamepokelewa`; the vendor gets `💳 Oda #... imelipwa kwa simu`; the rider is told not to take cash.
6. All money lands in **Mkoba's Selcom account**. `/admin/payments` shows what is owed to each vendor (paid orders) and each rider (paid fees for delivered jobs). The admin sends the money outside the system and clicks **Mark sent**.

Rules

- Payments are switched off until the Selcom keys are set; then no payment links are sent at all.
- `LIPA` only finds payments from the last 7 days; older unpaid ones are marked expired.

- Rejecting a paid order marks it "refund needed" for the admin; rejecting an unpaid one cancels the payment.
- Customers can always pay cash instead.

Weak points

- Mkoba holds other people's money and pays out by hand: a legal question and a trust and reconciliation burden.
- Refunds are manual and only noted, not tracked to completion.
- A paid order is not recorded as a sale until the vendor sends `maliza oda`.
- Outside the 21:00 check, a payment made without the notification arriving (and without the payer reopening the link) stays "pending" until evening.
- Never tested against Selcom's real system; the exact status words, PIN prompt behaviour across networks, and buyer-email requirement are unconfirmed.

18. Voice notes

Steps

1. Someone sends a WhatsApp voice note. Mkoba accepts it only from a vendor or household that has agreed, a rider, or a customer in an open shop visit.
2. Mkoba downloads the audio from WhatsApp and checks it is a WhatsApp voice note (Ogg/Opus) of **1 minute or less**.
3. It sends the audio to **Google Speech-to-Text** in Swahili (Tanzania), or English (US) for English users.
4. It replies `🎤 Nimesikia: "..."` with the words it heard.
5. The text is then handled **exactly like a typed message**, by the same rules and in the same order.
6. If the result is a record, it is saved as **pending** and Mkoba asks `NDIO` / `HAPANA`.

Rules

- Spoken quantities are understood (`kilo tano` → 5 kg).
- Errors get a plain message: too long, no words heard, unsupported audio, or try typing.
- Before Google is set up (or before consent), a voice note gets "I can't listen to voice notes yet, please type".

Weak points

- Only records wait for NDIO. A spoken **command** runs at once: a misheard `futa` deletes the last record, and so on.
- Swahili accuracy in noisy markets, with Sheng or mixed English, is unknown until tested.
- Every voice note costs money at Google, and the audio is sent to Google, likely outside Tanzania.
- Longer than 1 minute is refused outright rather than cut.

19. USSD

Steps

1. The vendor dials Mkoba's USSD code (to be leased through Africa's Talking). Africa's Talking sends each screen's choices to `mkoba.digital/ussd/<secret>`.
2. **New number:** a short data notice, `1. Nakubali / 2. Hapana`, then the name. The vendor is registered and told to dial again.
3. **Known vendor:** the menu: 1 sale, 2 purchase, 3 expense, 4 today, 5 this week, 6 debts, 7 stock, 8 delete last record.
4. **1–3:** asks the item (`0` to skip), then the amount; the record is saved **at once** and the screen says `Imehifadhiwa`.
5. **4–5:** sales, purchases, expenses and profit or loss.
6. **6:** up to 5 debts. **7:** up to 6 stock items, low ones marked `!`.
7. **8:** shows the newest record and asks `1. Ndio / 2. Hapana` before deleting it.

Rules

- Each screen is kept under 182 characters.
- Amounts must be digits (commas allowed), at least 2 digits.
- It is the same account as WhatsApp, in the vendor's language.
- Households and other kinds of number are told to use WhatsApp.

Differences

- No NDIO step for records, no quantities, no debt entry, no shop or bulk buying.
- Option 8 deletes one record, not the whole last message.

Weak points

- Not live: no shortcode is leased yet, and it is untested with Africa's Talking.

- A mistyped amount is saved without a chance to confirm.
- Using USSD triggers the evening-report bug (section 5).
- Each USSD session costs money; who pays is undecided.

20. Dashboard

Steps

1. The vendor sends `dashibodi` (also `dashboard`, `mtandaoni`).
2. Mkoba sends a one-time login link valid for **15 minutes**.
3. Opening the link shows an **Ingia** button; only pressing it logs in (so link-preview scanners can't use the link up).
4. The vendor stays logged in for **30 days** on that browser. **Toka** logs out.
5. The page shows totals for today, this week or this month (tabs), profit or loss for each of the last 14 days as a bar chart, recent records, debts, stock with low items, the credit score, and a button to open the Lite App.

Rules

- Only a hash of the link and of the session is stored.
- A used or expired link says so and tells the vendor to send `dashibodi` again.

Differences

- The dashboard is **read-only**; adding or changing anything is done on WhatsApp, the Lite App or USSD.

Weak points

- Swahili only, even for English vendors.
- Anyone holding the phone, or the unlocked browser, sees the vendor's business for 30 days.
- The link arrives by WhatsApp, so it only works within WhatsApp's window, which is fine since the vendor just asked.

21. Lite App

Steps

1. The vendor opens `mkoba.digital/d/app` (or the button on the dashboard) while logged in to the dashboard; the same 30-day login is used.

2. They can add it to the phone's home screen with **Weka Mkoba kwenye skrini ya simu** or the browser's **Add to Home screen**.
3. Five tabs, in the vendor's language:
   - **Nyumbani (Home):** totals for a chosen period;
   - **Rekodi (Records):** add a sale, purchase or expense with item, amount, and optional quantity and unit; recent records with a delete button;
   - **Stoo (Stock):** stock list and setting a count;
   - **Oda (Orders):** open and recent orders with accept, reject and complete;
   - **Madeni (Debts):** balances, and adding any of the four debt kinds by choosing it from a list.
4. **Offline:** the app itself loads from the phone. A record added without internet is kept on the phone and sent automatically when the connection returns. Data shown offline is the last copy downloaded.

Rules

- Every record carries an ID made on the phone, so a record sent twice (retry after a weak connection) is saved once.
- An offline record keeps the time it was typed, but never more than 7 days back.
- Records from the app are saved at once (no NDIO), since amounts are typed into a number field.
- Order buttons do exactly what the WhatsApp commands do (customer, rider and payment messages included).

Differences

- Debt kind is picked from a list, so the `nimelipa` / `nimemlipa` trap doesn't exist here.
- Any recent record can be deleted, not only the newest.

Weak points

- Unsent offline records live only in that browser's storage; clearing it, or a private window, loses them.
- Logging in still needs WhatsApp (`dashibodi`); the app can't log in by itself.
- Completing an order in the app shows the receipt as text, but there is no easy way to forward it from the app.
- Using the app triggers the evening-report bug (section 5).

22. Language

Steps

1. `english` (or `kiingereza`, `in english`) switches replies to English; `kiswahili` (or `swahili`) switches back. Mkoba confirms the switch.
2. Without a switch command, Mkoba changes language **only when a message is clearly one language**: it counts known English words and known Swahili words and verb forms, and switches when only one kind appears.
3. The choice is saved and used everywhere for that person: WhatsApp replies, evening reports, Lite App, USSD, voice recognition.

Rules

- A pasted mobile-money SMS never changes the language (providers write in English).
- Customers choose by the word they open a shop with (`duka` → Swahili, `shop` → English).

Weak points

- A vendor who types one English word (`sold 5000`) is switched to English without asking.
- English is missing some commands that Swahili has (paying back a debt) and English number words.
- The dashboard, web shop, payment pages and the evening-report template are Swahili only.

23. Privacy and data deletion

Steps

1. The privacy notice is at `mkoba.digital/faragha` (Swahili) and `/privacy` (English), linked from sign-up and shop visits.
2. The vendor sends `FUTA DATA` (also `futa taarifa`, `futa akaunti`, `delete my data`). Mkoba warns that everything will be deleted permanently and cannot be undone.
3. Within **10 minutes** the vendor sends `FUTA KABISA` (`delete everything`).
4. In one step Mkoba deletes the vendor and everything linked: records, debts, stock, receipts and points, shop items and orders, bulk lines, notices, statement links, lender permissions and views, scores, logins and all logged messages. Delivery jobs for them are removed. A group they created stays for the other members.

5. The reply says all data is deleted and backup copies delete themselves within 14 days.

Rules

- `FUTA DATA` works even before consent.
- `FUTA KABISA` without a recent request says there is no request waiting.
- Riders delete their own data with `futa data` (their open jobs go back to other riders).

Differences

- `futa` alone deletes only the last record; `FUTA DATA` deletes everything.

Weak points

- Payment records (Selcom) and orders the person placed **as a customer** at other shops are not deleted.
- Customers of a shop have no command to delete their own phone number and orders.
- Messages are stored for every number that writes, even before consent.
- Registration with Tanzania's data protection commission and rules on sending data abroad (Google, server location) are unresolved.

24. Admin pages

Steps

1. The admin opens `mkoba.digital/admin` and logs in with user `admin` and the `ADMIN_PASSWORD` from the server settings.
2. **Main page:** vendor counts and activity (7 and 30 days), messages today, sales in the last 30 days, a 14-day chart, the parser failures to review (messages not understood, rejected, or corrected), and the vendor list with last-seen time.
3. **`/admin/bulk`:** create and switch collection points on or off, set the wholesale price list, see open and closed group orders, press **Goods arrived**.
4. **`/admin/lenders`:** add lenders, set or reset their portal password, switch them off.
5. **`/admin/riders`:** approve or switch off riders, add riders, change the delivery-fee settings, see deliveries.
6. **`/admin/payments`:** money received, what is owed to each vendor and rider, **Mark sent** after paying them, and recent payments with notes such as "refund needed".

Rules

- Without `ADMIN_PASSWORD` set, every admin page returns "not found".

- 5 wrong passwords from one address lock it out for 15 minutes.
- Every admin form must come from the Mkoba site itself (protects against forged requests).

Weak points

- One shared password, no individual admin accounts, no two-step login, and no record of which admin did what.
- The lock-out counter is kept in each server process's memory; with several processes running, the real limit is higher, and it resets on restart.
- Payouts, refunds, bulk purchasing and rider approval all depend on someone doing them by hand on these pages.